

- Let V be a vector space over $\mathbb{R}$ and consider a function $\mathbf{x}, \mathbf{y} \mapsto \langle \mathbf{x}, \mathbf{y} \rangle$ that sends ordered pairs of vectors to real numbers. We call this an *inner product* if for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ and $a, b \in \mathbb{R}$ we have
 - $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$,
 - $\langle \mathbf{x} + \mathbf{y}, \mathbf{z} \rangle = \langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle$,
 - $\langle a\mathbf{x}, \mathbf{y} \rangle = a\langle \mathbf{x}, \mathbf{y} \rangle$,
 - $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$,
 - $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ implies $\mathbf{x} = \mathbf{0}$.

For the rest of the study sheet we assume that V is an inner product space.

- A subset $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ of V is called *orthogonal* if it doesn't contain the zero vector and if $\langle \mathbf{u}_i, \mathbf{u}_j \rangle = 0$ for all $i \neq j$. The set is called *orthonormal* if

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \begin{cases} 1 & i = j, \\ 0 & i \neq j. \end{cases}$$

- If $\mathbf{u}_1, \dots, \mathbf{u}_k$ is an orthogonal set and $\mathbf{x} = a_1\mathbf{u}_1 + \dots + a_k\mathbf{u}_k$ then

$$a_i = \frac{\langle \mathbf{x}, \mathbf{u}_i \rangle}{\langle \mathbf{u}_i, \mathbf{u}_i \rangle}.$$

- Orthogonal sets are independent.
- Let $U \subseteq V$ be a subspace and let $\mathbf{u}_1, \dots, \mathbf{u}_k$ be a basis for U . We define *orthogonal projection* $P_U : V \rightarrow V$ by

$$P_U(\mathbf{x}) = \frac{\langle \mathbf{x}, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1 + \frac{\langle \mathbf{x}, \mathbf{u}_2 \rangle}{\langle \mathbf{u}_2, \mathbf{u}_2 \rangle} \mathbf{u}_2 + \dots + \frac{\langle \mathbf{x}, \mathbf{u}_k \rangle}{\langle \mathbf{u}_k, \mathbf{u}_k \rangle} \mathbf{u}_k.$$

- We have $\langle \mathbf{x} - P_U(\mathbf{x}), \mathbf{u} \rangle = 0$ for all $\mathbf{u} \in U$. It follows from this and the Pythagorean theorem that

$$\|\mathbf{x} - \mathbf{u}\| \geq \|\mathbf{x} - P_U(\mathbf{x})\| \quad \text{for all } \mathbf{u} \in U.$$

- Let $U \subseteq V$ be a subspace. We define the *orthogonal complement* by

$$U^\perp := \{\mathbf{v} \in V : \langle \mathbf{v}, \mathbf{u} \rangle = 0 \text{ for all } \mathbf{u} \in U\}.$$

This is a subspace of V satisfying $U \cap U^\perp = \{\mathbf{0}\}$. If V is finite dimensional, we also have $V = U + U^\perp$. These two facts imply that any $\mathbf{v} \in V$ has a unique expression of the form $\mathbf{v} = \mathbf{u} + \mathbf{w}$ with $\mathbf{u} \in U$ and $\mathbf{w} \in U^\perp$. We summarize this by writing

$$V = U \oplus U^\perp.$$

- For any $\mathbf{v} \in V$, the unique pair of $\mathbf{u} \in U$ and $\mathbf{w} \in U^\perp$ satisfying $\mathbf{v} = \mathbf{u} + \mathbf{w}$ are the same as the orthogonal projections: $\mathbf{u} = P_U(\mathbf{v})$ and $\mathbf{w} = P_{U^\perp}(\mathbf{v})$.